

AutoML Insights: sample_churn.csv

AutoML Analysis & Business Insights Report

Prepared For: PredictIQ Enterprise User
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Platform: PredictIQ Studio AutoML Engine

1. Dataset Profile & Summary

The AutoML engine parsed the dataset **sample_churn.csv**. Below is an overview of the dataset dimensions and data quality metrics.

Metric	Value
Dataset Name	sample_churn.csv
Total Row Count	20
Total Column Count	7
File Size	2.8 KB
Data Quality Score	100/100

2. Applied Data Cleaning Steps

The following data cleansing and preprocessing pipeline steps were executed prior to training:

- Encode Categorical Columns
- Standardize Numeric Features
- Standardize Numeric Features

3. AutoML Model Comparison

A total split validation was run on multiple algorithms. The leaderboard below displays performance metrics. Algorithms are ranked based on their validation score.

Model	Accuracy	Precision	Recall	F1 Score	CV Score	Time
Logistic Regression	100.00%	100.00%	100.00%	100.00%	100.00%	0.013s
Decision Tree	100.00%	100.00%	100.00%	100.00%	100.00%	0.005s
Random Forest	100.00%	100.00%	100.00%	100.00%	100.00%	0.066s
Support Vector Machine	100.00%	100.00%	100.00%	100.00%	100.00%	0.017s
KNN	100.00%	100.00%	100.00%	100.00%	90.00%	0.003s
Naive Bayes	100.00%	100.00%	100.00%	100.00%	100.00%	0.001s

XGBoost	100.00%	100.00%	100.00%	100.00%	95.00%	1.840s
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4. Best Model Deep-Dive

The best performing model is **Logistic Regression**. It was selected automatically based on optimization targets for 'churned'.

5. AI Business Insights & Recommendations

This section outlines tactical business findings derived from model metrics, weights, and feature importance.

Executive Summary

We performed automated machine learning (AutoML) on the dataset 'sample_churn.csv' with the goal of predicting the target column 'churned'. The problem was identified as a classification task (predicting a categorical class). After evaluating multiple models, the Logistic Regression model was selected as the optimal predictor with a Accuracy of 100.0% and F1-score of 100.0%.

Data Quality Review

Quality Rating: Excellent (Score: 100/100). The dataset has minimal missing values and duplicates. Data cleaning steps successfully handled remaining discrepancies. Columns with high missing rates or high variance were normalized, and outliers were analyzed to improve stability.

Primary Business Drivers (Feature Importance)

The analysis indicates that the most influential features driving predictions are: contract_type, monthly_charges, satisfaction_score. These parameters hold the highest weights or split-gains, making them key leverage points for understanding model behavior.

Actionable Recommendations

- Focus business strategy adjustments around the primary drivers (contract_type, monthly_charges, satisfaction_score).
- Continuously monitor model performance as new data flows in to verify that 'churned' matches predictions.
- Implement data ingestion validation checks to maintain high data quality scores.

Model Strengths	Model Weaknesses
• Highly optimized preprocessing pipeline handled missing values and encoding automatically. • The Logistic Regression model is robust against overfitting and offers balanced metrics.	• Rule-based fallback was used instead of Gemini LLM. Standard summaries might lack contextual domain explanations. • If some features are highly correlated, individual feature importance could be slightly distributed.
Possible Bias & Risks	Suggestions for Improvement

- Class imbalance or distribution skew in target 'churned' can skew model outputs. Regular review is advised.

- Provide a Gemini API key in Settings to unlock deep, contextual generative business analysis.
- Add more domain-specific features or collect more rows to further boost validation metrics.